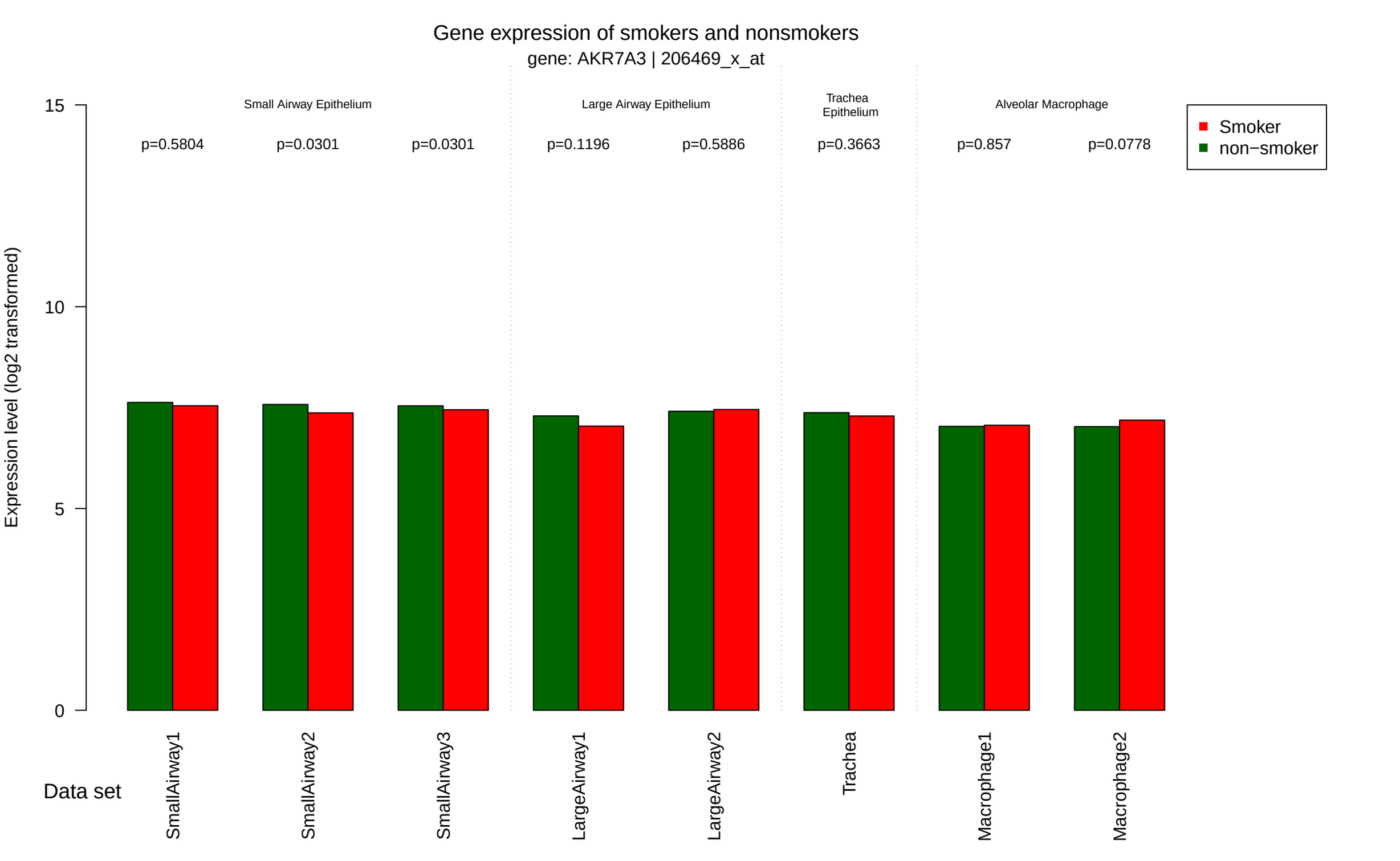
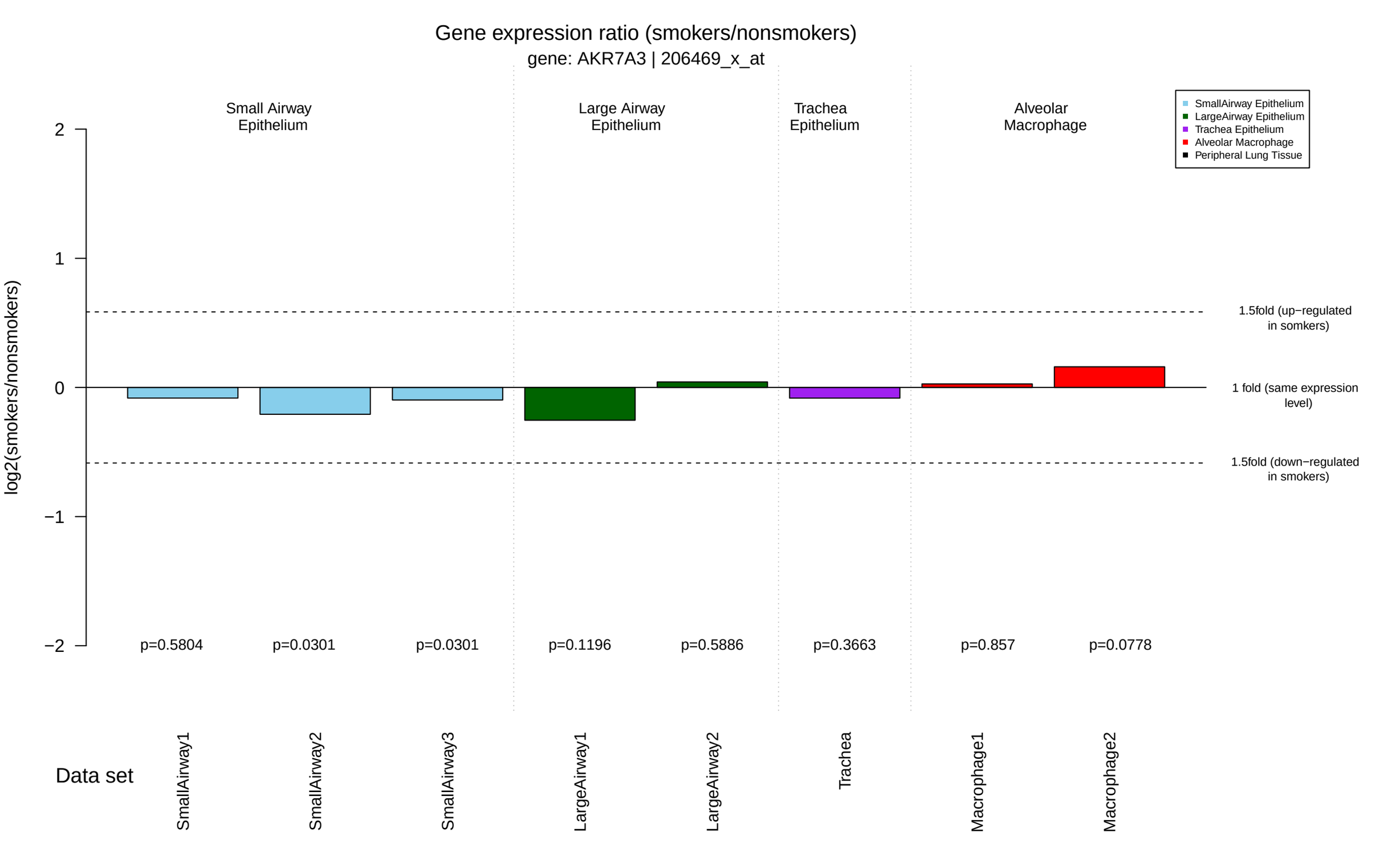
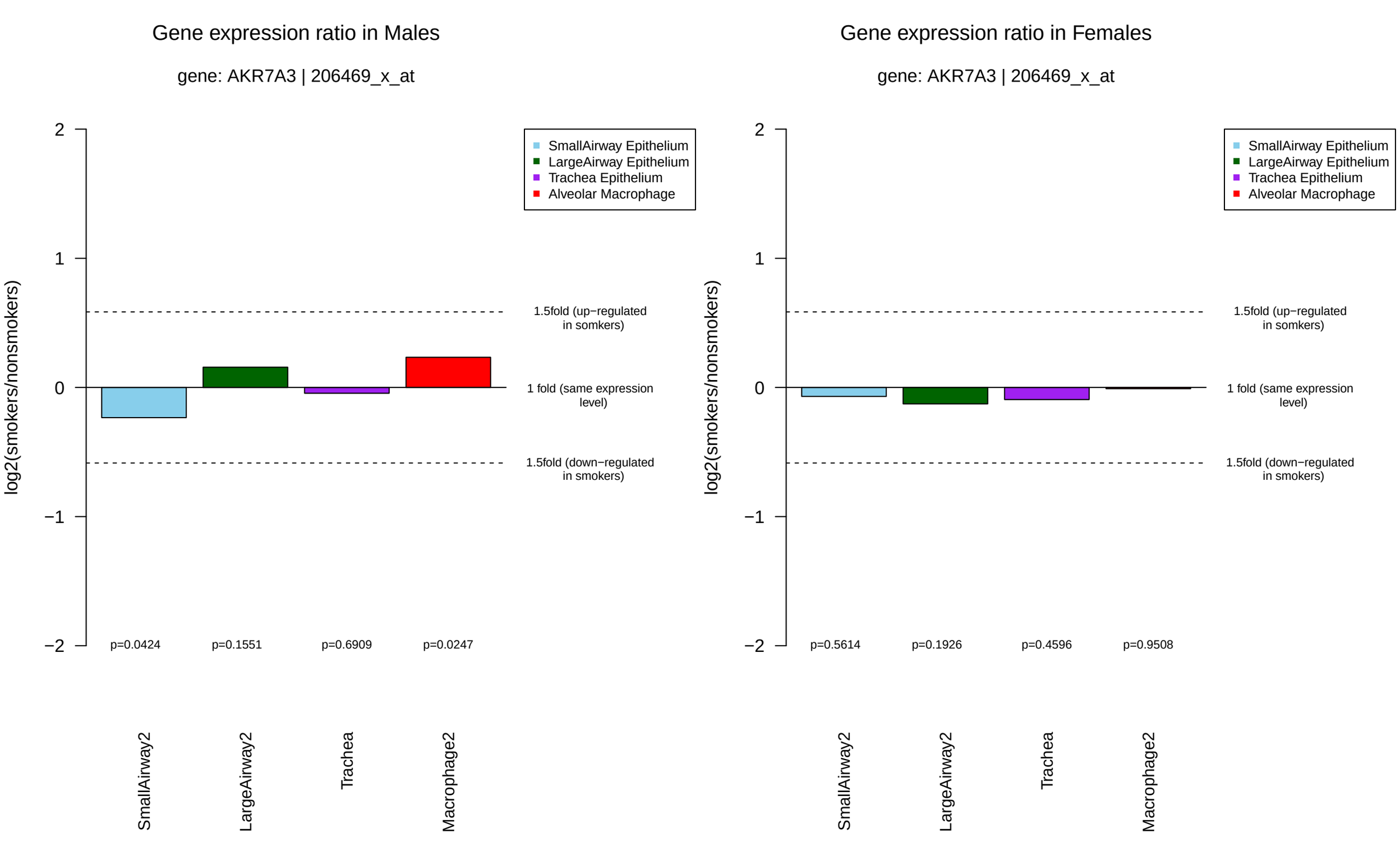
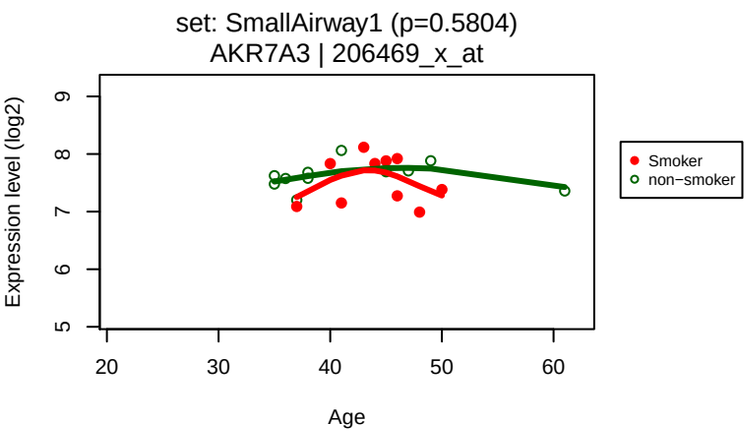
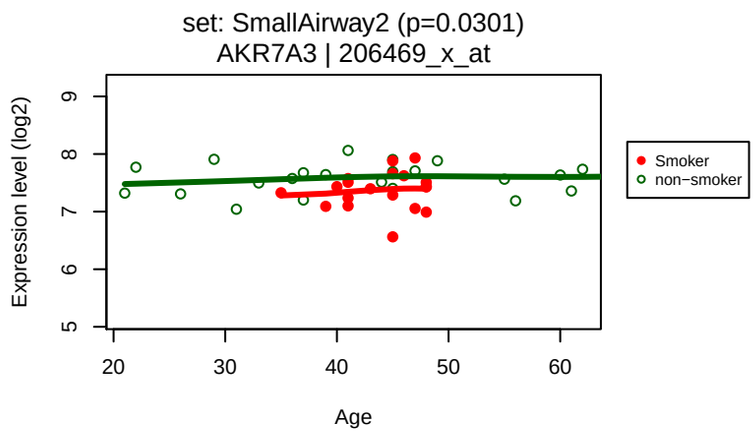
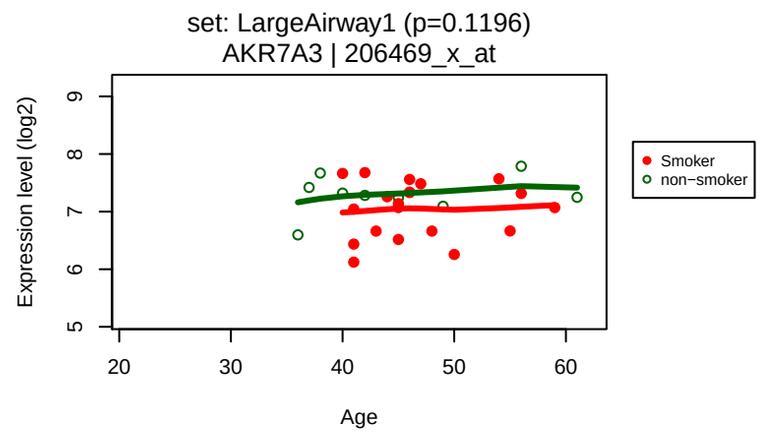

Gene expression of smokers and nonsmokers
gene: AKR7A3 | 206469_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5804
p=0.0301
p=0.0301
p=0.1196
p=0.5886
p=0.3663
p=0.857
p=0.0778
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

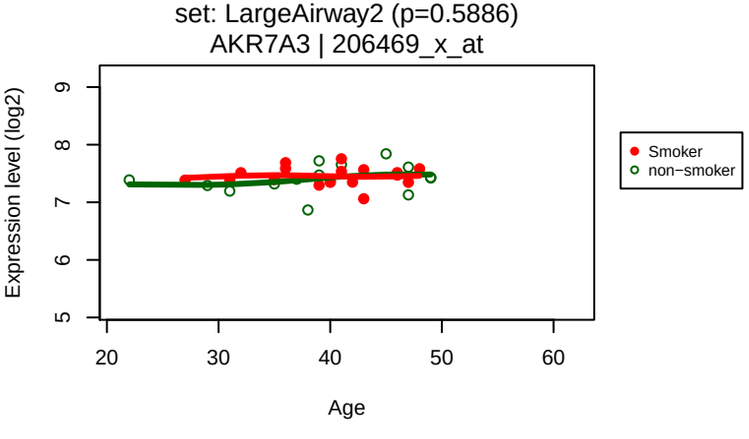

Gene expression ratio (smokers/nonsmokers)
gene: AKR7A3 | 206469_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5804
p=0.0301
p=0.0301
p=0.1196
p=0.5886
p=0.3663
p=0.857
p=0.0778
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

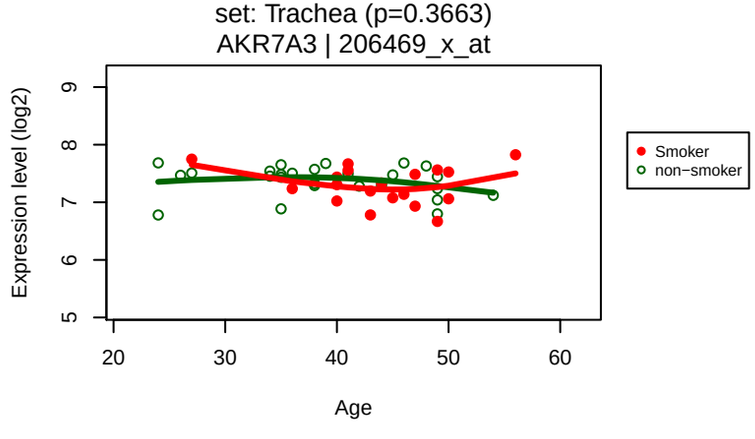

Gene expression ratio in Males
gene: AKR7A3 | 206469_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0424
p=0.1551
p=0.6909
p=0.0247
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: AKR7A3 | 206469_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5614
p=0.1926
p=0.4596
p=0.9508
SmallAirway2
LargeAirway2
Trachea
Macrophage2

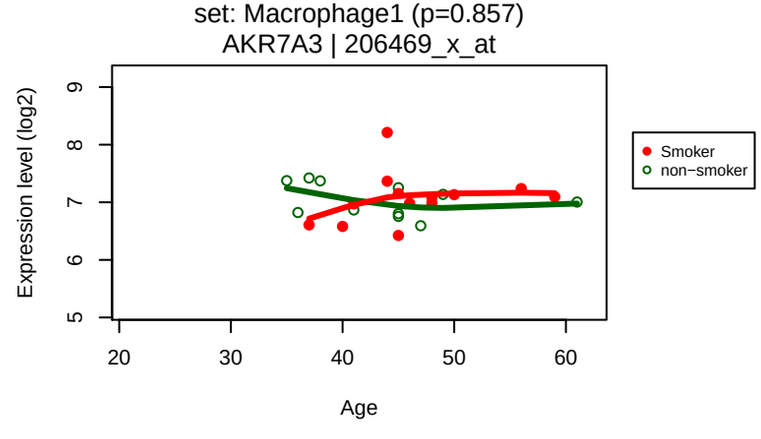

set: SmallAirway1 (p=0.5804)
AKR7A3 | 206469_x_at
Expression level (log2)
Age
Smoker
non−smoker

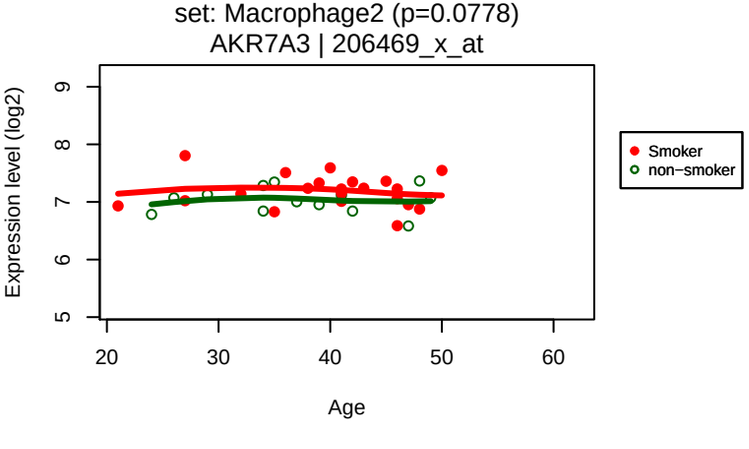

set: SmallAirway2 (p=0.0301)
AKR7A3 | 206469_x_at
Expression level (log2)
Age
Smoker
non−smoker

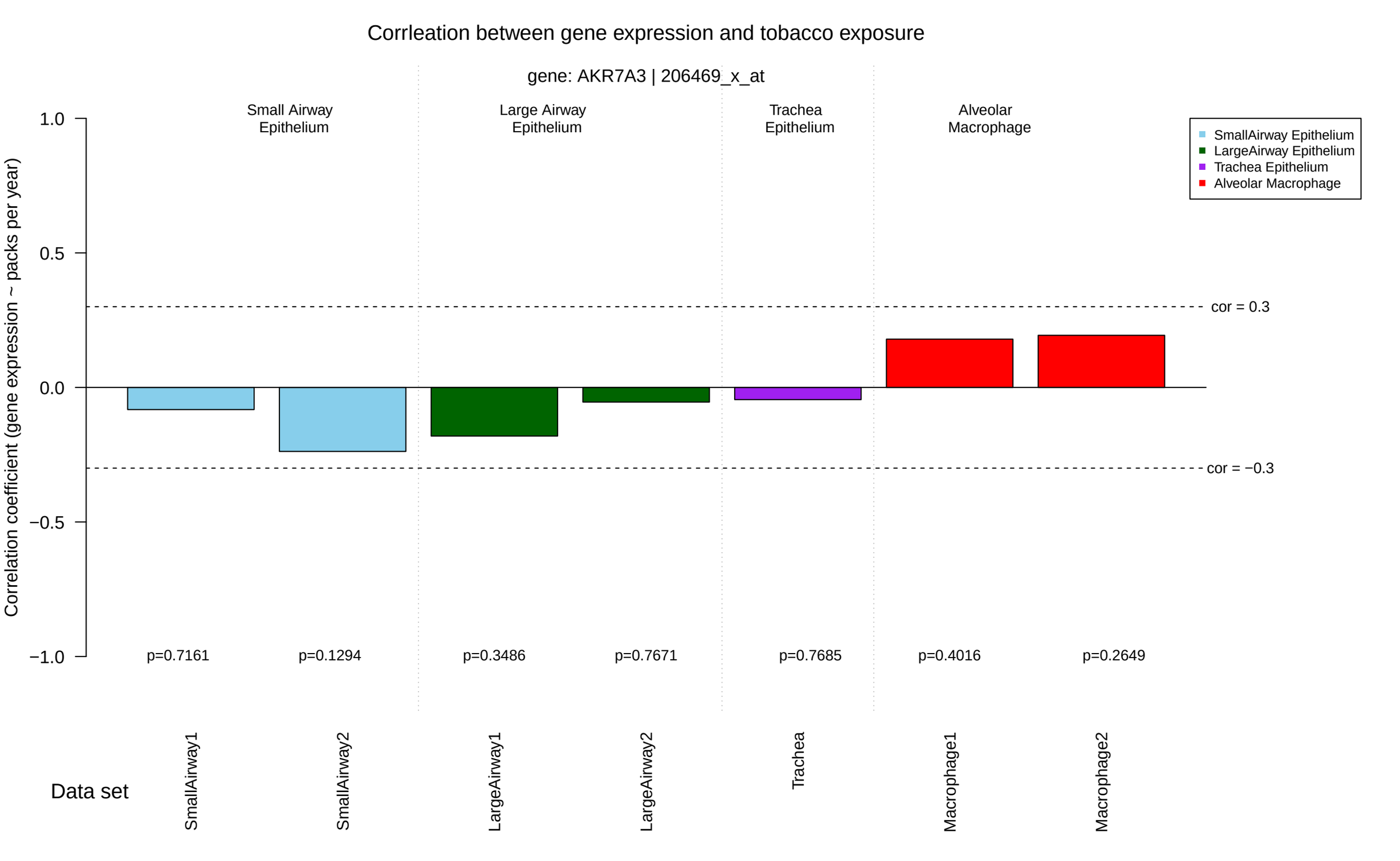

set: LargeAirway1 (p=0.1196)
AKR7A3 | 206469_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5886)
AKR7A3 | 206469_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3663)
AKR7A3 | 206469_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.857)
AKR7A3 | 206469_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0778)
AKR7A3 | 206469_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: AKR7A3 | 206469_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7161
p=0.1294
p=0.3486
p=0.7671
p=0.7685
p=0.4016
p=0.2649
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2